The Systems Behind Modern AI

Sam Ly
Professor Thanh Nguyen
CS 4630

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Abstract

Rendering complex graphics in real time requires millions of floating point operations, a task fundamentally unfit for CPU's. Pioneered by NVIDIA, GPUs were conceived in the late 20th century as a solution to this problem. Initially, the architecture of GPU's was justifiably rigid, with specific graphics rendering stages baked in as hardware primitives. Despite this, the graphics API's exploited to perform general purpose scientific compute (GPGPU). Demand for general purpose parallel processing on GPUs was one of the factors that led to a complete redesign of the device architecture. In 2006, NVIDIA released the G80 architecture, which replaced the graphics specific hardware stages with a unified pool of compute cores, known as Streaming Multiprocessors (SMs). Alongside this release, NVIDIA released the CUDA programming language in 2007, exposing these general parallel processors to programmers. This parallelism lends itself nicely to the computational primitives required in modern machine learning and artificial intelligence. Efficiently executing matrix products requires a deep exploration of the CUDA programming model and device architecture. Concretely, we introduce the Roofline model to describe the parameters needed for effective device utilization. Although GPUs have been the most popular option for AI compute, they are far from ideal. To conclude, we briefly discuss some alternative parallel processing architectures that have recently developed.